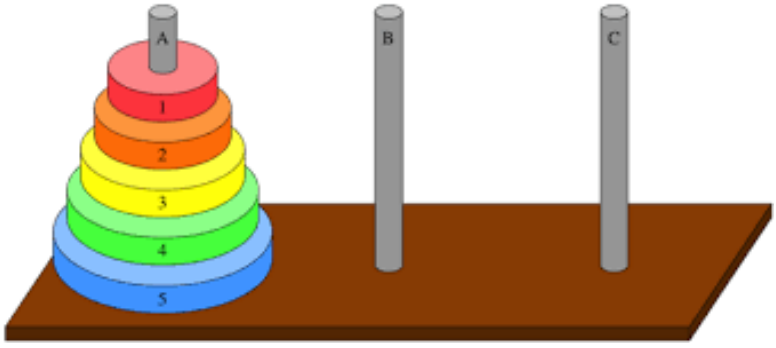
Assignment2 – Prolog

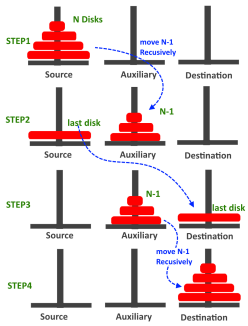
Problem 2.1 (Towers of Hanoi)

The Towers of Hanoi is a puzzle. It consists of three pegs (A, B, and C) and a number of disks of different sizes, which can slide onto any peg. The puzzle starts with the disks in a stack in ascending order of size on one peg, the smallest at the top, thus making a conical shape. The objective of the puzzle is to move all disks from peg A to peg B, while obeying the following rules:

1. only one disk can be moved at a time,
2. each move consists of taking the upper disk from one of the stacks and placing it on top of another stack or on an empty peg,
3. no larger disk may be placed on top of a smaller disk.

The idea of the algorithm (for $N > 1$) is to move the top $N - 1$ disks onto the auxiliary peg, then move the bottom disk to the destination peg, and finally moving the remaining $N - 1$ disks from the auxiliary peg to the destination peg.





1. Write a Prolog predicate that prints out a solution for the Towers of Hanoi puzzle. Use the `write(X)` predicate that prints the value of X (X can be simple text or any type of argument) to the screen and `nl` that prints a new line

to write a rule `move(N, A, B, C)` that prints out the solution for moving N disks from peg A to peg B, using C as the auxiliary peg.
Each step of the solution should be of the form "Move top disk from X to Y".
Examples:

```
?- write(hello), write('␣world!'), nl.
hello world!
true.
```

```
?- move(3, left, center, right).
Move top disk from left to center
Move top disk from left to right
Move top disk from center to right
Move top disk from left to center
Move top disk from right to left
Move top disk from right to center
Move top disk from left to center
true ;
false.
```

Solution:

```
move(1, A, B, _) :-
    write('Move␣top␣disk␣from␣'),
    write(A), write('␣to␣'), write(B), nl.
move(N, A, B, C) :-
    N>1, M is N-1,
    move(M, A, C, B), move(1, A, B, _), move(M, C, B, A).
```

---

2. Determine the complexity class of your algorithm in terms of the number of disks N and explain how you computed it.

Solution: Let $T(N)$ be the number of moves needed to move N disks from one peg to another.
Clearly, $T(1) = 1$. For $T(N)$, we have the following recursive relation:

$$T(N) = 2T(N-1) + 1$$

The values for $N = 1, 2, 3, 4, 5$ are 1, $2+1$, 2^2+2+1, 2^3+2^2+2+1, and $2^4+2^3+2^2+2+1$. Thus, $O(T(n)) = 2^{n-1}$, which is exponential.
(You could also solve the non-homogenous linear recurrence to obtain a precise closed formula for $T(N)$.)
